

Midterm 1 Comprehensive Study Guide — Stat 490

Design of Experiments — Spring 2026

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1. Foundations: Hypothesis Testing & Two-Sample Inference

1.1 Core Definitions

- **Population:** Full set of units. **Sample:** Subset used for inference. **SRS:** Every unit has equal chance.
- H_0 (null): No effect/difference. H_a (alternative): What we seek evidence for.
- **Type I Error** (α): Reject H_0 when true (false positive). **Type II** (β): Fail to reject H_0 when H_a true (false negative).
- **p-value:** $P(\text{test stat as extreme or more} \mid H_0 \text{ true})$. Reject H_0 if p-value $< \alpha$.
- **Power** $= 1 - \beta = P(\text{Reject } H_0 \mid H_a \text{ true})$. Increases with $\uparrow n$, $\uparrow$ effect size, $\uparrow \alpha$, $\downarrow \sigma$.

1.2 t-Tests

Assumptions: (1) Random sampling, (2) Approx. normality (critical for small n), (3) Independence between groups. *One-sample t:* SRS from population; errors $X - \mu$ i.i.d; population approximately normal or n large.

Two-sample pooled t: two independent SRSs; both populations approximately normal; equal variances ($\sigma_1^2 = \sigma_2^2$); can check via SD ratio, F-test, Bartlett, or Levene.

Two-sample Welch t: two independent SRSs; normal or large n ; variances may differ, but distributions should have similar shape.

Paired t: pairs form an SRS of differences; differences d_i approximately normal or n large; pairs independent of each other.

Scenario	Test Statistic	df
One-sample	$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$	$n - 1$
Pooled two-sample	$t = (\bar{x}_1 - \bar{x}_2) / (s_p \sqrt{1/n_1 + 1/n_2})$	$n_1 + n_2 - 2$
Welch (non-pooled)	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{s_1^2/n_1 + s_2^2/n_2}$	Satterthwaite
Paired	$t = \bar{d} / (s_d / \sqrt{n_d})$	$n_d - 1$

Theory: every t statistic has the form

$$t = \frac{\text{signal}}{\text{noise}} = \frac{\text{estimated effect}}{\text{standard error under } H_0},$$

e.g. $(\bar{x} - \mu_0) / (s / \sqrt{n})$ or $(\bar{x}_1 - \bar{x}_2) / SE$. Under H_0 we plug the random sample SD s in place of σ , which makes the standardized statistic follow a t distribution instead of a normal one.

Pooled vs. Welch: Pool when SD ratio in $[1/2, 2]$; else use Welch/Satterthwaite. **Paired:** Natural pairs (same subject, before/after). Reduces variability—extension of blocking.

Satterthwaite: Approximation for degrees of freedom in Welch's t-test: $\Delta = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$.

TI-84: STAT $\rightarrow$ TESTS $\rightarrow$ 2:T-Test (one-sample), 4:2-SampTTest (two-sample). For pooled: set Pooled:Yes. Enter $\bar{x}$, s , n directly via "Stats" input.

1.3 CIs and Assumption Checks

CI for μ_i : $\bar{x} \pm t_{\alpha/2, df} \cdot s / \sqrt{n}$. CI for $\mu_1 - \mu_2$: $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df} \cdot SE$. If CI excludes 0 $\Rightarrow$ reject H_0 .

- **Normality:** QQ-plot (straight line = normal), Shapiro-Wilk test (highest power for small n).
- **Equal variance (k groups):** **Bartlett's**—powerful if normal, sensitive to non-normality. **Levene's**—robust (uses median absolute deviations). Rule: max SD $> 2 \times$ min SD $\Rightarrow$ suspect.

2. One-Way ANOVA (Fixed Effects, CRD)

2.1 The Model

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \dots, a, \quad j = 1, \dots, n, \quad \varepsilon_{ij} \sim \text{NID}(0, \sigma^2)$$

μ : grand mean. τ_i : treatment effect, $\sum \tau_i = 0$. Treatment mean $\mu_i = \mu + \tau_i$. $N = an$ (balanced).

Fixed-effects one-factor ANOVA assumptions:

- Errors ε_{ij} are independent (ensured by randomization).
- $\varepsilon_{ij} \sim N(0, \sigma^2)$ for all i, j (same distribution and variance across groups).
- The factor is correctly specified (no hidden time trends or other structure left in the errors).

Three assumptions: (1) Independence of errors (via randomization). (2) Normality of errors. (3) Constant variance σ^2 across all groups (homoscedasticity).

2.2 ANOVA Decomposition and Table

Key identity: $SS_T = SS_{\text{Treat}} + SS_E$. Total variability = between-group + within-group.

$$SS_T = \sum_{i,j} (y_{ij} - \bar{y}_{..})^2, \quad SS_{\text{Treat}} = \sum_i n(\bar{y}_{i.} - \bar{y}_{..})^2, \quad SS_E = \sum_{i,j} (y_{ij} - \bar{y}_{i.})^2$$

Under $H_0 : \mu_1 = \dots = \mu_a$,

$$E[MS_{\text{Treat}}] = \sigma^2, \quad E[MS_E] = \sigma^2,$$

so the F statistic MS_{Treat}/MS_E should be near 1. Under H_a (at least one μ_i differs),

$$E[MS_{\text{Treat}}] = \sigma^2 + \frac{n \sum_{i=1}^a \tau_i^2}{a-1} > \sigma^2,$$

so F tends to be larger than 1. Intuitively, F is “between-group variance estimate / within-group variance estimate”.

Robustness: the ANOVA F -test is fairly robust to mild non-normality, especially for balanced designs and moderate sample sizes; violations of equal variances are usually more serious than modest departures from normality.

Source	df	SS	MS	F
Treatment	$a-1$	SS_{Trt}	$MS_{\text{Trt}} = SS_{\text{Trt}}/(a-1)$	MS_{Trt}/MS_E
Error	$N-a$	SS_E	$MS_E = SS_E/(N-a)$	
Total	$N-1$	SS_T		

Why it works: Under H_0 , both MS_{Trt} and MS_E estimate σ^2 , so $F \approx 1$. Under H_a , $E[MS_{\text{Trt}}] = \sigma^2 + n \sum \tau_i^2/(a-1) > \sigma^2 = E[MS_E]$, so F is large. MS_E is always an unbiased estimator of σ^2 .

Estimates: $\hat{\mu} = \bar{y}_{..}$, $\hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$, $\hat{\mu}_i = \bar{y}_{i.}$. Residual: $e_{ij} = y_{ij} - \bar{y}_{i.}$.

$R^2 = SS_{\text{Trt}}/SS_T$ = proportion of variability explained by treatment.

2.3 Regression Equivalence

ANOVA with a levels $\equiv$ regression with $a-1$ dummy variables. Same F , MS_E , p-value. Regression $\hat{\beta}_j$ = difference from base level; ANOVA $\hat{\tau}_i$ = deviation from grand mean.

2.4 CIs from ANOVA

- For μ_i : $\bar{y}_{i.} \pm t_{\alpha/2, N-a} \sqrt{MS_E/n}$
- For $\mu_i - \mu_j$: $(\bar{y}_{i.} - \bar{y}_{j.}) \pm t_{\alpha/2, N-a} \sqrt{2 MS_E/n}$

2.5 Worked Example: Dog Food Shelf Height (CRD)

3 shelf heights (Knee, Eye, Waist), $n = 8$ per group, $N = 24$.

	Knee	Eye	Waist
$\bar{y}_{i.}$	81.0	84.625	90.875

Grand mean $\bar{y}_{..} = 85.5$. From ANOVA: $SS_{\text{Trt}} = 399.2$, $SS_E = 288.7$, $MS_E = 13.75$.

$F = 199.62/13.75 = 14.52$ with df (2, 21). p-value = 0.00011 $\Rightarrow$ reject H_0 at $\alpha = 0.05$.

95% CI for μ_{Knee} : $81 \pm 2.08 \sqrt{13.75/8} = 81 \pm 2.73 = (78.27, 83.73)$.

TI-84 for F p-value: 2nd $\rightarrow$ DISTR $\rightarrow$ Fcdf(14.52, 1E99, 2, 21) returns ≈ 0.00011 .
TI-84 for t critical value: 2nd $\rightarrow$ DISTR $\rightarrow$ invT(0.025, 21) returns -2.080 .
SS by hand: Enter group data in L1, L2, L3; use 1-Var Stats L1 to get $\bar{x}$, $\sum x^2$, n per group.

3. Multiple Comparisons

3.1 The Familywise Error Problem

The definition of the “family” depends on context: it might be all pairwise comparisons, a small set of pre-planned contrasts, or all contrasts you decide to examine after looking at the data.

With $\binom{a}{2}$ pairwise tests at α : FW error = $1 - (1 - \alpha)^{\binom{a}{2}}$. For $a = 4$: $K = 6$ tests, FW error ≈ 0.26 at $\alpha = 0.05$. Each additional test inflates the chance of a false positive.

3.2 Named Procedures

Method	Formula / Key Idea	When to Use	FW?
Fisher LSD	$LSD = t_{\alpha/2, N-a} \sqrt{2MS_E/n}$. Declare sig. if $ \bar{y}_{i.} - \bar{y}_{j.} > LSD$.	After sig. F ; few groups. Most liberal.	No
Bonferroni	Replace α with α/K in each CI/test.	Small K ; any comparisons. Conservative.	Yes
Tukey HSD	$HSD = q_{\alpha, a, N-a} \sqrt{MS_E/n}$. Uses Studentized Range.	All pairwise. Less conservative than Bonferroni.	Yes
Dunnett	Compare each trt to one control only ($a-1$ tests). Uses special table.	Placebo/control design. Narrower CIs than Tukey.	Yes
Scheffé	Reject if $\frac{ \hat{\Gamma} /SE(\hat{\Gamma})}{\sqrt{(a-1)F_{\alpha, a-1, N-a}}} >$	Any contrast, post-hoc. Protects against data snooping.	Yes

Fisher LSD:

- Requires a significant overall ANOVA F first.
- Assumes the same ANOVA assumptions (normality, equal variance).
- Does *not* control familywise error well when many comparisons are made; relatively liberal.

Bonferroni:

- No special assumptions beyond those of the underlying t -tests.
- Uses α/K per test (or $1 - \alpha/K$ CIs) to keep FW error $\leq \alpha$ for any chosen family.

Tukey HSD:

- Based on the Studentized range distribution (max–min of sample means).
- Assumes equal variances and works best with balanced or nearly balanced designs.
- Designed for all pairwise comparisons with strong FW error control.

Dunnett:

- Similar spirit to Tukey but conditional on a single control mean.
- Uses the joint distribution of treatment–control differences.
- Only valid for comparing each treatment to one control (not all pairwise).

Scheffé:

- Uses a scaled F -statistic valid for *any* contrast, including those chosen after seeing the data.
- Guarantees FW error $\leq \alpha$ over all contrasts, but is very conservative, especially for simple pairwise differences.

Power ranking (pairwise): Fisher LSD > Tukey > Bonferroni > Scheffé.
For complex contrasts: Scheffé > Tukey. **For control comparisons only:** Dunnett > all others.

Optimal control allocation: With a treatments + 1 control, allocate $n_0 = n\sqrt{a}$ to control.

3.3 Worked Example: Dog Food Pairwise Comparisons

From §2.5: $a = 3$, $N = 24$, $MS_E = 13.75$, $n = 8$, error df = 21.

$SE = \sqrt{2(13.75)/8} = 1.854$

Pair	$ \bar{y}_{i.} - \bar{y}_{j.} $	LSD = 3.86	Tukey HSD = 4.67	Bonf. ($\alpha^*=0.0167$)
W–K	9.875	Sig	Sig	Sig
W–E	6.250	Sig	Sig	Sig
K–E	3.625	Not sig	Not sig	Not sig

LSD: $t_{0.025,21} \cdot 1.854 = 2.08 \cdot 1.854 = 3.86$. Tukey: $q_{0.05,3,21} \cdot \sqrt{13.75/8} = 3.56 \cdot 1.31 = 4.67$.
Bonferroni: $t_{0.0083,21} \cdot 1.854 = 2.62 \cdot 1.854 = 4.86$ (wider than Tukey—more conservative).

All three methods agree here, but for borderline cases the order matters.

TI-84: To get $t_{0.0083,21}$: `invT(0.0083, 21)` ≈ -2.62 . For F critical value: `invF(area, df1, df2)` is not built-in; instead compute p-value with `Fcdf`.

3.4 Contrasts and Orthogonal Contrasts

A **contrast**: $\Gamma = \sum c_i \mu_i$ where $\sum c_i = 0$. Pairwise diff is simplest contrast.

Examples: Placebo(1) vs avg of Trts 2,3: $C = (-2, 1, 1)$. Effect of protein P1 vs P2 in 2×2 factorial: $C = (0.5, 0.5, -0.5, -0.5)$.

Two contrasts C, D are **orthogonal** if $\sum c_i d_i = 0$. There are exactly $a - 1$ mutually orthogonal contrasts. They are independent and partition SS_{Trt} into single-df components.

Example (3 treatments): $C_1 = (-2, 1, 1)$ and $C_2 = (0, -1, 1)$ are orthogonal: $(-2)(0) + (1)(-1) + (1)(1) = 0$. Note: $(1, -1, 0)$ and $(0, 1, -1)$ are NOT orthogonal.

$SS_{\text{contrast}} = \frac{n(\sum c_i \bar{y}_{i.})^2}{\sum c_i^2}$ with 1 df. Test: $F = SS_{\text{contrast}}/MS_E$.

3.5 Data Snooping and Scheffé

Data snooping: Picking contrasts AFTER seeing data inflates Type I error (you implicitly tested many). **Scheffé protects:** guarantees FW error $\leq \alpha$ for *all possible* contrasts, even those chosen post-hoc. More conservative than Tukey for pairwise, but necessary for arbitrary contrasts.

4. Model Adequacy & Transformations

4.1 Residual Diagnostics

Residual $e_{ij} = y_{ij} - \bar{y}_{i.}$. Checks:

- **Normality:** QQ-plot of residuals (straight line = good). Shapiro–Wilk test.

- **Constant variance:** Residuals vs. fitted. Funnel/fan $\Rightarrow$ heteroscedasticity. Bartlett/Levene test.
- **Outliers:** Studentized residual $|e_{ij}^*| > 3 \Rightarrow$ suspect outlier.
- **Independence:** Ensured by randomization. Plot residuals vs. run order for time effects.

4.2 Variance-Stabilizing Transformations

Theory (delta method idea): if $\text{Var}(Y) \approx g(\mu)$ and $h'(\mu) \propto 1/\sqrt{g(\mu)}$, then $\text{Var}(h(Y))$ is approximately constant, which motivates choices like $h(y) = \sqrt{y}$ when $\text{Var}(Y) \propto \mu$ (Poisson) or $h(y) = \log y$ when $\text{Var}(Y) \propto \mu^2$.

Practical decision rule:

- If variance increases with the mean $\Rightarrow$ try $\sqrt{\cdot}$ or $\log(\cdot)$.
- If response is a proportion near 0 or 1 $\Rightarrow$ try $\arcsin \sqrt{\cdot}$.
- If transformations fail and residuals show strong non-normality or outliers, consider non-parametric tests instead of t/F .

Distribution	$\text{Var}(Y) \propto$	Transform	Example
Poisson (counts)	μ	$\sqrt{Y}$	Flaw counts by coating
Binomial (proportions)	$\mu(1 - \mu)$	$\sin^{-1}(\sqrt{Y})$	Proportion of defectives
Lognormal/Multiplicative	μ^2	$\ln(Y)$	Prices, concentrations

Worked example: Copper wire flaws (3 coatings, $n = 10$ each). Before transform: funnel-shaped residual plot, $MS_E = 8.97$. After $\sqrt{Y}$ transform: residuals uniform, $MS_E = 0.232$. Both reject H_0 , but transformed model satisfies assumptions.

TI-84: To transform data: store in L1, then $L2 = \sqrt{(L1)}$. Re-run **1-Var Stats** on L2.

5. Random Effects Model

5.1 Fixed vs. Random: When to Use

Fixed: The a levels are the *specific* ones of interest. Inference is about those levels only.
Random: The a levels are a *random sample* from a larger population. Inference is about the population.

Examples: 3 specific labs $\Rightarrow$ fixed. 3 labs sampled from all labs $\Rightarrow$ random. 5 specific therapists $\Rightarrow$ fixed. 5 therapists randomly chosen $\Rightarrow$ random.

Random one-way effects assumptions:

- Levels $i = 1, \dots, a$ are a random sample from a larger population of levels.
- $\tau_i \sim N(0, \sigma_\tau^2)$ i.i.d.
- $\varepsilon_{ij} \sim N(0, \sigma^2)$, independent of each other and of all τ_i .
- Within each level i , the Y_{ij} are exchangeable (same distribution for all j at that level).

5.2 The Model and Expected Mean Squares

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \tau_i \sim N(0, \sigma_\tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2), \quad \text{independent}$$

Total variance: $\text{Var}(Y_{ij}) = \sigma_\tau^2 + \sigma^2$. Within-treatment correlation: $\text{Cov}(Y_{ij}, Y_{ij'}) = \sigma_\tau^2$.

Expected mean squares—the central result:

$$E[MS_{\text{Trt}}] = \sigma^2 + n\sigma_\tau^2, \quad E[MS_E] = \sigma^2$$

Under $H_0 : \sigma_\tau^2 = 0$, both estimate σ^2 , so $F = MS_{\text{Trt}}/MS_E \approx 1$. Same F -test as fixed effects.

5.3 Key Differences from Fixed Effects

Aspect	Fixed	Random
Hypothesis	$H_0 : \tau_1 = \dots = \tau_a = 0$	$H_0 : \sigma_\tau^2 = 0$
$E[MS_{\text{Trt}}]$	$\sigma^2 + \frac{n \sum \tau_i^2}{a-1}$	$\sigma^2 + n\sigma_\tau^2$
Within-trt correlation	0	$\sigma_\tau^2 > 0$
CI for μ uses	MS_E/N	MS_{Trt}/N (wider!)
Inference scope	Specific levels	Population of levels

Why is random-effects CI wider? Because the treatment effect itself is a source of randomness. Fixed model treats treatments as deterministic.

Heuristic derivation: MS_{Trt} averages squared deviations of group means from the grand mean. Each group mean has variance σ^2/n from noise plus σ_τ^2 from the random effects, so

$$E[MS_{\text{Trt}}] = \sigma^2 + n\sigma_\tau^2, \quad E[MS_E] = \sigma^2.$$

The F -test MS_{Trt}/MS_E therefore tests $H_0 : \sigma_\tau^2 = 0$ (no between-level variance).

ICC interpretation: $\rho = \sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$ is the correlation between two units from the same level, and the fraction of total variance explained by “which level you are in”. If $\rho \approx 0$, grouping is not important; if $\rho \approx 1$, observations within a level are almost identical.

5.4 Variance Components and ICC

Estimators: $\hat{\sigma}^2 = MS_E$. $\hat{\sigma}_\tau^2 = (MS_{\text{Trt}} - MS_E)/n$.

Negative $\hat{\sigma}_\tau^2$: Can happen if true σ_τ^2 is small, n is small, or there's negative within-group correlation the model doesn't capture. Interpret as $\sigma_\tau^2 \approx 0$.

Intraclass Correlation (ICC):

$$\rho = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} = \text{Corr}(Y_{ij}, Y_{ij'})$$

Proportion of total variance due to between-group differences. $\rho = 0$: no group structure. $\rho \rightarrow 1$: within-group nearly identical. Exact CI via F -distribution.

5.5 Worked Example: Sire-Calf Inheritance

5 sires (random sample from population), 8 calves each, $N = 40$.

From ANOVA: $MS_{\text{Trt}} = 1397.8$, $MS_E = 463.8$, $F = 3.014$, $p = 0.031$.

$\hat{\sigma}^2 = 463.8$. $\hat{\sigma}_\tau^2 = (1397.8 - 463.8)/8 = 116.75$.

$\text{ICC} = 116.75/(116.75 + 463.8) = 0.201$.

Interpretation: About 20% of the total variability in calf weight is due to differences between sires. Most variability is within-sire.

CI for μ (random): $82.55 \pm t_{0.025,4} \sqrt{MS_{\text{Trt}}/N} = 82.55 \pm 2.04 \sqrt{1397.8/40} = (70.5, 94.6)$.

CI for μ (fixed): $82.55 \pm t_{0.025,35} \sqrt{MS_E/N} = (75.6, 89.5)$. Random CI is wider.

TI-84: F p-value: $\text{Fcdf}(3.014, 1\text{E}99, 4, 35) \approx 0.031$. For χ^2 CI on σ^2 : $\chi^2\text{cdf}$ is available.

6. RCBD (Blocking)

6.1 Motivation and Additive Model

Blocking: Group units into homogeneous blocks to reduce error. "Block what you can, randomize what you cannot." RCBD = every treatment appears once in every block. Extension of paired t -test to $a > 2$.

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad i = 1, \dots, a, \quad j = 1, \dots, b, \quad \sum \tau_i = 0, \quad \sum \beta_j = 0$$

RCBD assumptions:

- Within each treatment-block cell, errors ε_{ij} are independent across all i, j .
- Errors are approximately normal with common variance σ^2 .
- Additive model: no treatment $\times$ block interaction; the difference between any two treatments is the same in every block.

How to detect interaction: Residuals vs. blocks plot shows patterns. But with one obs per cell, you *cannot* formally test for interaction.

6.2 RCBD ANOVA Table

Source	df	SS	F
Treatment	$a - 1$	SS_{Trt}	MS_{Trt}/MS_E
Block	$b - 1$	SS_{Block}	MS_{Block}/MS_E
Error	$(a - 1)(b - 1)$	SS_E	
Total	$N - 1$	SS_T	

$N = ab$. Key: Error df = $(a - 1)(b - 1)$, not $N - a$!

For mixed RCBD with random blocks,

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \beta_j \sim N(0, \sigma_\beta^2),$$

with treatments fixed. Expected mean squares:

$$E[MS_{\text{Trt}}] = \sigma^2 + \frac{b \sum \tau_i^2}{a - 1}, \quad E[MS_{\text{Block}}] = \sigma^2 + a \sigma_\beta^2, \quad E[MS_E] = \sigma^2.$$

If a real treatment $\times$ block interaction exists but is not modeled, its variation is absorbed into MS_E , inflating the error term and reducing power for both treatment and block F -tests (higher Type II error).

6.3 Degrees of Freedom: CRD vs. RCBD

	CRD	RCBD
Error df	$N - a = a(n - 1)$	$(a - 1)(b - 1)$
Example: $a = 4$, $b = 3$ blocks	$12 - 4 = 8$ (if $n = 3$)	$(3)(2) = 6$

RCBD has fewer error df, but typically MS_E is much smaller because block variability is removed. Net effect: usually more power.

6.4 Why Blocking Helps (Numerical)

Apple orchard: 5 treatments, 6 locations (blocks).

	With blocking (RCBD)	Without blocking (CRD)
MS_E	34,455 ($\sqrt{} \approx 186$)	52,938 ($\sqrt{} \approx 230$)
Treatment p-value	0.682	0.823

Blocking reduced MS_E by 35%. Same conclusion here (not significant), but in cases where treatment effects are moderate, this reduction would flip the result.

6.5 Multiple Comparisons in RCBD

Same methods (Tukey, Bonferroni, LSD, etc.) with modifications:

- Replace n (reps per treatment) with b (number of blocks).
- Error df = $(a - 1)(b - 1)$.
- $SE = \sqrt{2 MS_E / b}$.

Tukey HSD for apple data: $SE = \sqrt{34455/6} = 75.78$. $q_{0.05,5,20} = 4.23$. $HSD = 4.23 \times 75.78 = 320.6$.

6.6 Mixed Model: Fixed Treatments, Random Blocks

When blocks are a random sample (e.g., operators, days, subjects):

- ANOVA table, df, and F -test for treatments **unchanged**.
- Block variance: $\hat{\sigma}_B^2 = (MS_{\text{Block}} - MS_E)/a$.
- Focus of inference: **treatments** (fixed). Blocks just model extra variability.

Assembly time example: 4 machines (fixed), 6 operators (random). $MS_{\text{Machine}} = 5.308$, $MS_E = 1.590$. $F = 3.34$, $p = 0.048$. Machines are significantly different. $\hat{\sigma}_{\text{Operator}}^2 = (8.417 - 1.590)/4 = 1.707$.

7. Power & Sample Size

7.1 Conceptual Framework

Power = $P(\text{Reject } H_0 \mid H_a \text{ true})$. Steps: (1) Find rejection region under H_0 . (2) Compute probability of landing there under specific H_a .

What increases required n ? Higher desired power, smaller effect size, smaller α , larger σ^2 .
What decreases required n ? Larger effect size, larger α , smaller σ^2 .

7.2 Power for Z and t Tests

One-sample example: $H_0 : \mu = 368$ vs $H_a : \mu > 368$. $n = 50$, $s = 15$, $\bar{x} = 372.5$.

Rejection region at $\alpha = 0.05$: reject if $\bar{X} > 368 + 1.645 \cdot (15/\sqrt{50}) = 371.5$.

Power at $\mu = 375$: $P(\bar{X} > 371.5 \mid \mu = 375) = P\left(Z > \frac{371.5 - 375}{15/\sqrt{50}}\right) = P(Z > -1.65) = 0.95$.

Interpretation: 95% chance of correctly detecting the difference if true mean is 375.

7.3 Effect Size for ANOVA (Cohen's f)

$$f = \frac{\sigma_{\text{between}}}{\sigma_{\text{within}}} = \sqrt{\frac{\frac{1}{a} \sum (\mu_i - \bar{\mu})^2}{\sigma^2}}$$

Benchmarks: $f = 0.10$ (small), $f = 0.25$ (medium), $f = 0.40$ (large).

7.4 Worked Example: 4-Group Power

$H_a : \mu_1 = 50, \mu_2 = 60, \mu_3 = 50, \mu_4 = 60$. $\sigma^2 = 25$.

$\bar{\mu} = 55$. Between-group var = $\frac{1}{4}[(50 - 55)^2 + (60 - 55)^2 + (50 - 55)^2 + (60 - 55)^2] = 25$.

$f = \sqrt{25/25} = 1.0$ (very large). For power ≥ 0.90 at $\alpha = 0.05$: $n \approx 3$ per group.

If σ increases to 6: $f = \sqrt{25/36} = 0.833$, need larger n . If $\sigma = 7$: $f = 0.714$, even larger n .

Key insight: Doubling σ roughly quadruples the required n (since $n \propto 1/f^2 \propto \sigma^2$).

TI-84: For the power curve, compute the rejection threshold under H_0 using `invNorm` or `invT`, then find the probability under H_a using `normalcdf` or `tcdf` with the shifted mean.

7.5 CI-Based Sample Size

To estimate $\mu_i - \mu_j$ within margin ME at $100(1 - \alpha)\%$: solve $ME = t_{\alpha/2, N-a} \sqrt{2 MS_E / n}$ iteratively (since $N = na$ depends on n). Start with a guess, compute, refine.

8. Non-Parametric Alternatives

Parametric vs non-parametric assumptions:

- Wilcoxon signed-rank vs paired *t*: requires that the distribution of differences is symmetric about the median; no normality needed. More robust to outliers; slightly less powerful than *t* if data truly normal.
- Mann–Whitney vs two-sample *t*: independent samples, similar shape distributions; tests a shift in location (e.g. $P(X > Y) \neq 0.5$). If shapes differ, it is closer to testing “stochastic ordering”.
- Kruskal–Wallis vs ANOVA *F*: independent samples and continuous distributions; null is equal distributions across groups, and the test is based on rank sums. Under true normality with equal variances, *F* is more powerful; under heavy tails or strong skew, Kruskal–Wallis *H* can be preferable.

8.1 When to Use

When normality fails and no transformation works. Trade-off: fewer assumptions but lower power under normality. Generally based on ranks rather than raw values.

Parametric	Non-Parametric	Assumption	Tests
Paired <i>t</i>	Sign test	None	Median diff $\neq 0$
Paired <i>t</i>	Wilcoxon signed-rank	Symmetric dist	Median/location shift
Two-sample <i>t</i>	Mann–Whitney (rank-sum)	Similar shapes	Location shift
ANOVA <i>F</i>	Kruskal–Wallis <i>H</i>	Continuous data	Equal distributions

8.2 Wilcoxon Signed-Rank Test

For testing median: $H_0 : \tilde{\mu} = m_0$.

Procedure: (1) Compute $d_i = x_i - m_0$. Drop zeros. (2) Rank $|d_i|$ from smallest to largest. (3) Attach signs. (4) $W^+ =$ sum of positive ranks.

Under H_0 : $E(W^+) = \frac{n(n+1)}{4}$, $\text{Var}(W^+) = \frac{n(n+1)(2n+1)}{24}$.

For moderate n : $Z = (W^+ - E(W^+))/\sqrt{\text{Var}(W^+)}$ is approx. $N(0, 1)$.

Mini-example: Weekly food costs, $n = 10$, $H_0 : \tilde{\mu} = 157$, $H_a : \tilde{\mu} < 157$.

$W^+ = 9$. $E(W^+) = 10(11)/4 = 27.5$. $SD = \sqrt{10(11)(21)/24} = 9.81$.

$Z = (9 - 27.5)/9.81 = -1.89$. p-value ≈ 0.03 . Reject H_0 —evidence median < 157 .

8.3 Mann–Whitney (Rank-Sum) Test

Two independent groups (m and n observations). (1) Pool and rank all $m + n$ values. (2) $W_1 =$ sum of ranks in group 1. (3) $U_1 = W_1 - m(m + 1)/2$.

$E(U_1) = mn/2$, $\text{Var}(U_1) = mn(m + n + 1)/12$.

Mini-example: Learning time with/without experience. $m = 7$ (no exp), $n = 8$ (exp).

$W_1 = 74$ (no-exp ranks). $U_1 = 74 - 7(8)/2 = 46$. $E = 28$, $SD = 8.64$.

$Z = (46 - 28)/8.64 = 2.08$. p ≈ 0.02 . Inexperienced workers take longer.

8.4 Kruskal–Wallis *H* Test

Non-parametric one-way ANOVA. (1) Rank all N observations jointly. (2) $T_i =$ rank sum for group i .

$$H = \frac{12}{N(N + 1)} \sum_{i=1}^a \frac{T_i^2}{n_i} - 3(N + 1)$$

Under H_0 : $H \sim \chi^2_{a-1}$ approximately. Large $H \Rightarrow$ reject.

Power comparison: Under normality, ANOVA *F* is more powerful than K-W. Under heavy non-normality or outliers, K-W can be preferable.

TI-84: For χ^2 p-value: $\chi^2\text{cdf}(H, 1E99, a-1)$. For normal approx in signed-rank/Mann-Whitney: use `normalcdf`.

9. Experimental Design Principles

9.1 Three Principles

- **Randomization:** Randomly assign treatments to units $\Rightarrow$ eliminates systematic bias, ensures independence. Without it, you can only show association, not causation.
- **Replication:** Multiple observations per treatment $\Rightarrow$ estimate error variance, improve precision. $\neq$ repeated measures on the same unit (that’s pseudo-replication).
- **Blocking:** Group homogeneous units $\Rightarrow$ reduce MS_E , increase power. Block on known nuisance factors.

Example: Testing 3 fertilizers on crop yield. *Randomization:* randomly assign fertilizers to plots. *Replication:* use 10 plots per fertilizer. *Blocking:* if field has north/south moisture gradient, block by location.

9.2 Design Comparison

Design	Nuisance	Model	Error df
CRD	None	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	$N - a$
RCBD	1 block factor	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	$(a - 1)(b - 1)$
Latin Sq	2 block factors	$y_{ijk} = \mu + \tau_j + \rho_i + \gamma_k + \varepsilon$	$(p - 2)(p - 1)$

9.3 Quick Concept Checks

- ANOVA treats quantitative and qualitative factors the same for SS computation. (**True**—interpretation differs, formulas don't.)
- Fisher's LSD is extremely conservative. (**False**—it's the most liberal.)
- Tukey's method compares all treatments to a control. (**False**—that's Dunnett's.)
- MS_E estimates σ^2 (the error variance). (**True**.)
- In mixed RCBD (random blocks), primary inference is on treatments. (**True**—blocks just capture nuisance variability.)
- Blocking always reduces error df. (**True**—but typically reduces MS_E more, giving net power gain.)
- Kruskal-Wallis is always more powerful than ANOVA F . (**False**—less powerful under normality.)

10. Common Distributions Reference

Distribution	Mean	Variance	Use / Context
$N(\mu, \sigma^2)$	μ	σ^2	Error terms, CLT
t_ν	0	$\nu/(\nu - 2)$	Small-sample means
F_{d_1, d_2}	$d_2/(d_2 - 2)$	complex	ANOVA, variance ratios
χ^2_ν	ν	2ν	Variance CIs, K-W test
Poisson(λ)	λ	λ	Counts; $\sqrt{Y}$ transform
Binomial(n, p)	np	$np(1 - p)$	Proportions; $\sin^{-1}\sqrt{Y}$

TI-84 Distribution Functions (all under 2nd → DISTR):
normalcdf(lo, hi, μ , σ) — normal prob. invNorm(area, μ , σ) — normal quantile.
tcdf(lo, hi, df) — t prob. invT(area, df) — t quantile.
Fcdf(lo, hi, df1, df2) — F prob. χ^2 cdf(lo, hi, df) — χ^2 prob.
Tip: For upper-tail p-value of $F = 14.52$ with (2,21): Fcdf(14.52, 1E99, 2, 21).
Tip: For 1-Var Stats on lists: STAT → CALC → 1-Var Stats L1. Gives $\bar{x}, \sum x, \sum x^2, s, n$.

Leave space below for your own textbook examples and additional notes.